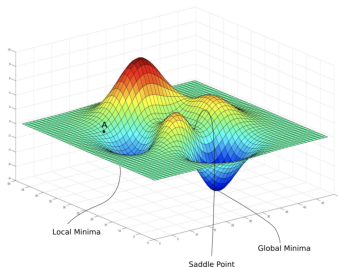
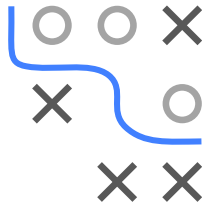


Optimization in Machine Learning

Foundations

Conditions for optimality

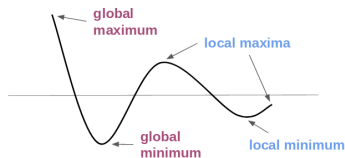
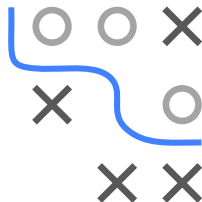


Learning goals

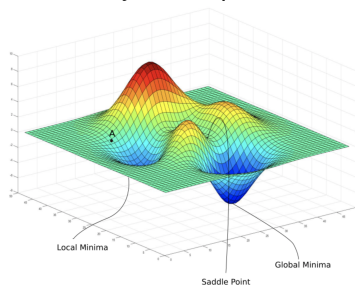
- Local and global optima
- Strict, isolated, and unique minimizers
- First & second order conditions

EXTREMA AND SADDLE POINTS

- Given $\mathcal{S} \subseteq \mathbb{R}^d$, $f : \mathcal{S} \rightarrow \mathbb{R}$
- Global minimum at $\mathbf{x}^*$: $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \mathcal{S}$
- Local minimum at $\mathbf{x}^*$: $\exists \epsilon > 0$ s.t. $f(\mathbf{x}^*) \leq f(\mathbf{x})$ for all $\mathbf{x} \in \mathcal{S} \cap B_\epsilon(\mathbf{x}^*)$ (ϵ -ball)
- Analogously for global and local max
- We call $\mathbf{x}^*$ saddle point if in feasible portion of every eps-ball $\mathcal{S} \cap B_\epsilon(\mathbf{x}^*)$, is at least a strictly better and a strictly worse point



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VARIANTS OF MINIMIZERS

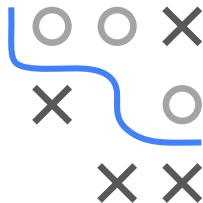
- Several refined notions for $\mathbf{x}^* \in \mathcal{S}$:

local min: $f(\mathbf{x}^*) < f(\mathbf{x})$ for all $\mathbf{x} \in \mathcal{S} \cap B_\epsilon(\mathbf{x}^*)$ with $\mathbf{x} \neq \mathbf{x}^*$

isolated local min: $\mathbf{x}^*$ is a local min and $B_\epsilon(\mathbf{x}^*)$ contains no other local min

global min: $\mathbf{x}^*$ is the only global minimizer on $\mathcal{S}$

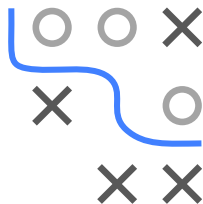
- Hierarchy: unique global $\Rightarrow$ isolated local $\Rightarrow$ strict local $\Rightarrow$ local
- Reverse implications generally fail
- **Caveat:** strict local does NOT imply isolated – example
 $f(x) = x^4(2 + \sin(1/x))$ at $x = 0$



EXISTENCE OF OPTIMA

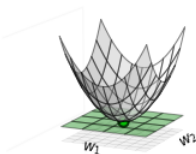
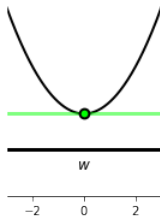
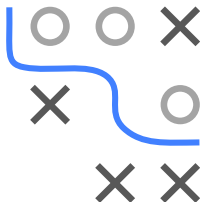
- $f : \mathcal{S} \rightarrow \mathbb{R}$
- If f continuous and $\mathcal{S}$ compact: minimum and maximum exist (extreme value theorem)
- If f discontinuous: no general existence statement
- Negative example, with $\mathcal{S} = [0, 1]$:

$$f(x) = \begin{cases} 1/x & x > 0 \\ 0 & x = 0 \\ -1/x & x < 0 \end{cases}$$



FIRST-ORDER CONDITION

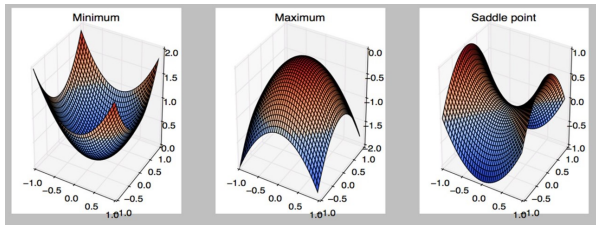
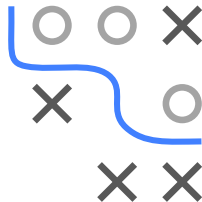
- Let $f : \mathcal{S} \rightarrow \mathbb{R}$, f differentiable, $\mathbf{x}^*$ interior point of $\mathcal{S}$
- Necessary condition:
If $\mathbf{x}^*$ is a local extremum, then $\nabla f(\mathbf{x}^*) = 0$
- Such points are called “stationary”
- Intuition: at a local extremum, the function must be flat, otherwise we can find a direction to move to a better value
- Not sufficient, e.g. saddle points are possible



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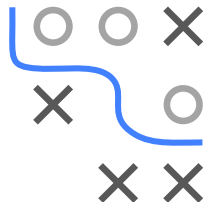
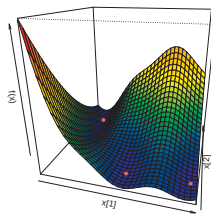
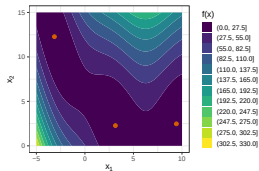
SECOND-ORDER CONDITION

- Let $f : \mathcal{S} \rightarrow \mathbb{R}$, $f \in \mathcal{C}^2$, $\mathbf{x}^*$ interior point of $\mathcal{S}$
- If $H(\mathbf{x}^*)$ is definite, then $\mathbf{x}^*$ is a strict local extremum
- If $H(\mathbf{x}^*)$ is semi-definite, then $\mathbf{x}^*$ is a local extremum
- If $H(\mathbf{x}^*)$ is indefinite, then $\mathbf{x}^*$ is a saddle point
- If $H(\mathbf{x}^*)$ is p(s)d, then $\mathbf{x}^*$ is a (strict) local min
this implies f is locally (strictly) convex
- If $H(\mathbf{x}^*)$ is n(s)d, then $\mathbf{x}^*$ is a (strict) local max
this implies f is locally (strictly) concave
- Interpretation: curvature pos (or neg) in all directions



EXAMPLE: BRANIN FUNCTION

- Branin function with 3 local minima

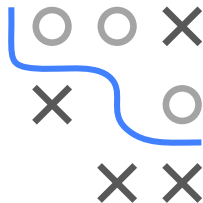


- EVs of Hessian at local minima:

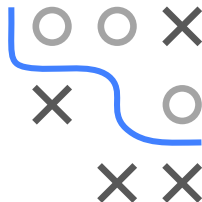
	λ_1	λ_2
Left	22.29	0.96
Middle	11.07	1.73
Right	11.33	1.69

CONVEXITY AND OPTIMA

- $f : \mathcal{S} \rightarrow \mathbb{R}$ convex on convex set $\mathcal{S}$
- Any local minimum is global
- The set of minima is convex
- If f strictly convex: at most one local minimum (unique global on $\mathcal{S}$, if it exists)
- Analogously for concave functions



CONVEXITY AND OPTIMA: PROOFS



1 = global. Assume $\mathbf{x}^*$ local min but not global, so $\exists \bar{\mathbf{x}} \in \mathcal{S}$ with $f(\bar{\mathbf{x}}) < f(\mathbf{x}^*)$. By convexity, $\forall t \in [0, 1]$:

$$f(\mathbf{x}^* + t(\bar{\mathbf{x}} - \mathbf{x}^*)) \leq (1 - t)f(\mathbf{x}^*) + tf(\bar{\mathbf{x}}) < f(\mathbf{x}^*)$$

For t small enough, the point $\mathbf{x}^* + t(\bar{\mathbf{x}} - \mathbf{x}^*) \in B_\epsilon(\mathbf{x}^*) \cap \mathcal{S}$ – contradiction.

2 = not convex. Let $\mathbf{x}^*, \bar{\mathbf{x}}$ both be global mins, so $f(\mathbf{x}^*) = f(\bar{\mathbf{x}}) = f^*$. By convexity, $\forall t \in [0, 1]$:

$$f(\mathbf{x}^* + t(\bar{\mathbf{x}} - \mathbf{x}^*)) \leq (1 - t)f^* + tf^* = f^*$$

Since f^* is the global min, equality must hold, so $\mathbf{x}^* + t(\bar{\mathbf{x}} - \mathbf{x}^*)$ is also a global minimizer.

3 = not one min. Same setup with $\mathbf{x}^* \neq \bar{\mathbf{x}}$, both global mins. Strict convexity gives **strict** inequality $f(\mathbf{x}^* + t(\bar{\mathbf{x}} - \mathbf{x}^*)) < f^*$ for $t \in (0, 1)$ – contradiction.

EXAMPLE

- $f(x, y) = x^4 + y^4 - x^2 - y^2$
- $\nabla f(x, y) = (4x^3 - 2x, 4y^3 - 2y)$
- $H(x, y) = \begin{pmatrix} 12x^2 - 2 & 0 \\ 0 & 12y^2 - 2 \end{pmatrix}$
- At $(0, 0)^T$ we have strict local max
- At $(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}})^T$ we have 4 strict local min
- At $(0, \pm \frac{1}{\sqrt{2}})^T, (\pm \frac{1}{\sqrt{2}}, 0)^T$ we have 4 saddle points

